

## 2MVA Level I Multivariable & Vector Analysis (25667)

### ASSESSED PROBLEM SHEET 4

The submission deadline is 5:00pm on Wednesday 10 December 2025  
(Answer all Questions)

Question 1: Let  $V$  be the volume given by  $0 \leq x \leq 1$ ,  $0 \leq y \leq 1$  and  $0 \leq z \leq 1$ . Consider the vector field  $\mathbf{F}$  in  $\mathbb{R}^3$  given by  $\mathbf{F} = (4xz, -y^2, yz)$ . Calculate  $\iiint_V \nabla \cdot \mathbf{F} dV$ .

Question 2: Let  $S$  be the surface along  $z = x^2 + y^2$  which lies below the plane  $2x + 2y + z = 2$ . Let  $C$  be the bounding curve of  $S$  which is traversed in the counter clockwise direction when looking from above. Consider the vector field  $\mathbf{F}$  in  $\mathbb{R}^3$  given by  $\mathbf{F} = (x + y, -y, 1)$ . Calculate  $\oint_C \mathbf{F} \cdot d\mathbf{r}$ .

Question 3: Let  $S$  be the surface of a Torus which is parameterized by the position vector  $\mathbf{r}(u, v)$  given by

$$\mathbf{r}(u, v) = (\cos(u)(R + r \cos(v)), \sin(u)(R + r \cos(v)), r \sin(v))$$

for  $0 \leq u \leq 2\pi, 0 \leq v \leq 2\pi$ .

- (a) Calculate the vectors  $\frac{\partial \mathbf{r}}{\partial u} = \left( \frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right)$  and  $\frac{\partial \mathbf{r}}{\partial v} = \left( \frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right)$ .
- (b) Now calculate the differential surface area element  $dS = \left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right| du dv$ .
- (c) Hence, by performing surface integration show that the surface area of the torus is given by

$$\iint_S dS = 4\pi^2 r R.$$